

Mini Project 2 Report

Analyzing Driver Behavior with RNN (LSTM)

1. Introduction

Ensuring road safety remains a major global priority, and evaluating how individuals drive can significantly lower the rate of traffic accidents. Thanks to the widespread use of modern smartphone sensors, tracking driving habits in real-time is now highly achievable. The primary goal of this project is to categorize driver actions utilizing sensor telemetry and to produce an ongoing driving score designed to boost overall safety.

2. Objectives

- Gather and format initial sensor telemetry, specifically from the gyroscope and accelerometer.
- Construct a time-series dataset leveraging the sliding window approach.
- Deploy a Long Short-Term Memory (LSTM) network to classify different driving behaviors.
- Produce an ongoing driver score alongside a detailed assessment report.
- Assess the reliability and overall performance of the machine learning model.

3. Dataset Description

The utilized data encompasses several driving sessions, with each log holding roughly five minutes of telemetry information.

- **Accelerometer Data:** Tracks acceleration components via X_Acc, Y_Acc, and Z_Acc.
- **Gyroscope Data:** Captures angular velocity components through X_Gyro, Y_Gyro, and Z_Gyro.

- **Quality Rating:** Evaluates overall driving performance using a standard 1 to 5-star metric.

4. Data Preprocessing

- **Sequence Formatting:** The raw telemetry was structured into sequential blocks, enabling the machine learning model to grasp the broader context of various driving maneuvers.
- **Data Normalization:** All sensor metrics were scaled to guarantee uniformity and handle differing measurement ranges effectively.

5. Model Architecture

This implementation relies on a Sequential LSTM network constructed using the TensorFlow/Keras framework. Because LSTM networks excel at identifying temporal dependencies within time-series inputs, they are perfectly suited for detecting specific driving habits across a given timeframe.

6. Training and Performance

Model training was conducted over 10 epochs utilizing a T4 GPU.

- **Initial Accuracy:** 76.99% at the first epoch.
- **Ending Loss:** 0.2125.
- **Final Accuracy:** The network reached an ultimate validation accuracy of 90.77%.

7. Driver Evaluation System

The proposed solution outputs a live "Driver Evaluation Report" by analyzing rolling windows of approximately 2 seconds (which translates to 20 time-steps).

Driver Status Classification	Detailed Description
Excellent Driver	Driving is safe and smooth (Score ≥ 80)
Good Driver	Slight aggression noticed (Score ≥ 60)
Average Driver	Performance requires improvement (Score ≥ 40)
Risky Driver	Elevated levels of jerky or rash movements (Score < 40)

8. Challenges & Data Constraints

- **Interference in Sensors:** Unprocessed inputs from gyroscopes and accelerometers frequently include high-frequency distortions due to rough terrain or engine tremors. Overcoming this necessitated strict preprocessing to guarantee the network trained on genuine vehicle maneuvers instead of background interference.
- **Smartphone Orientation:** The physical positioning of the recording device presents a major hurdle. Should the smartphone move while driving, its internal coordinate framework shifts. This project mitigated the issue by prioritizing relative shifts in movement over strict absolute coordinates.
- **Imbalanced Classes:** Under normal driving conditions, "Safe" behavior heavily outnumbers "Risky" incidents. A primary objective during model training was making sure the network could still correctly pinpoint these infrequent yet hazardous "High Risk" occurrences.

9. Future Scope & Enhancements

- **GPS Synchronization:** Merging GPS coordinates with sensor telemetry would allow the system to link rash driving events to contextual road environments, such as dense traffic, school districts, or tight corners.

- **Mobile App Integration:** The existing Python architecture could be ported to TensorFlow Lite for local execution within a mobile application. This enhancement would provide immediate audio warnings to users as soon as "Risky" actions are flagged.
- **Multi-Modal Systems:** Upcoming iterations might incorporate computer vision via camera streams to watch for driver distraction (like smartphone usage) or fatigue, combining visual cues with sensor readings for comprehensive 360-degree safety monitoring.
- **Cloud Analytics Dashboard:** Developing a centralized web hub would allow fleet administrators to concurrently monitor the scores of various drivers, yielding valuable performance trends and long-term analytics.

10. Project Hardware & Software Specifications

- **Software Stack:** The project utilizes Python 3.12, Pandas and NumPy for managing data, Matplotlib for graphing, alongside TensorFlow/Keras for the deep learning components.
- **Hardware Setup:** Training the LSTM sequences was accelerated using a Google Colab environment equipped with a Tesla T4 GPU, featuring 16GB of GDDR6 VRAM.

11. Conclusion

This initiative successfully proves that deep learning frameworks, notably LSTMs, are highly capable of processing sensor telemetry to forecast driving habits, achieving a strong accuracy rate of roughly 91%. Ultimately, this architecture holds significant potential for real-time driver coaching, overall safety tracking, and assessing insurance risk.